

CeresFlow Land Analysis Report

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Target Location:	prayagraj
Center Coordinates:	25.43813, 81.83380
Projected UTM Zone:	44N (EPSG:32644)
Bounding Box limits:	Lats: 25.30316 to 25.57310 Lons: 81.68459 to 81.98301

Summary Land Metrics

Index / Feature	Average Value	Target Range
Crop Health (NDVI)	0.500	0.0 (Bare) to 1.0 (Dense)
Moisture Index (NDWI)	-0.432	-1.0 (Dry) to 1.0 (Wet)
Mean Slope (Degrees)	0.5°	0° to 90°
Maximum Slope (Degrees)	1.1°	Risk threshold: > 15°
Soil Clay Index (SOCl)	-0.09	-1.0 (Sandy) to 1.0 (Clay)
Land Surface Temp (LST)	28.0°C	Ambient surface temperature proxy
Canopy Cover Fraction (CCF)	53.7%	0% to 100% field coverage

Farmer Recommendations & Interpretations

Crop Health Status:	Moderate / Fair Moderate vegetative growth. Some fields show sparse patches or sub-optimal plant density. <i>Action: Conduct a ground walk to check for localized nitrogen deficiency, early weed growth, or pests.</i>
Water/Moisture Status:	Water Deficit (Dry) High spectral dryness index. Soil is parched and foliage is exhibiting moisture stress. <i>Action: Irrigate affected sections immediately. Consider mulching to conserve moisture.</i>
Terrain Erosion Status:	Gentle / Safe Terrain Flat to gently rolling terrain (Max slope: 1.1°). Safe runoff speeds. <i>Action: Safe for standard row cultivation, mechanized seeders, and standard tractors.</i>

Soil Structure:	Loam / Balanced Soil Balanced Soil Index (Index: -0.09). Excellent water-holding capacity and fertility structure. <i>Action: Standard farming practices. Maintain organic matter inputs via crop rotation.</i>
Thermal Stress:	Optimal Thermal Zone Temperatures are within healthy crop growing parameters (Mean: 28.0°C). <i>Action: Ideal transpiration rates. No immediate temperature mitigation required.</i>
Canopy Spacing:	Moderate Canopy Cover Vegetation covers 53.7% of the field. Normal row-crop spacing pattern. <i>Action: Maintain 0.75m crop row interval spacing. Monitor weeds in unshaded row gaps.</i>

■ Scientific Procedure & Audit Log

This report was compiled automatically using Sentinel-2 bottom-of-atmosphere (Level-2A) spectral channels and SRTM Digital Elevation Models.

Mathematical Formula Definitions:

- *NDVI (Normalized Difference Vegetation Index)*: calculated as $(B08 - B04) / (B08 + B04)$, measuring leaf chlorophyll density.
- *NDWI (Normalized Difference Water Index)*: calculated as $(B03 - B08) / (B03 + B08)$, measuring foliage moisture.
- *SOCI (Soil Clay Index)*: calculated as $(Red - Green) / (Red + Green)$ to proxy mineral fraction.
- *LST (Land Surface Temp)*: calculated as $38.0 - 18.0 * NDVI - 0.0065 * DEM$ anomaly temperature model.
- *CCF (Canopy Cover Fraction)*: calculated by scaling NDVI from bare earth (0.15) to full canopy (0.80).
- *Slope Calculation*: Horn's 3x3 kernel convolution algorithm was used to calculate rise over run in local UTM coordinates.
- *Contour Lines*: Topographic contour lines generated at 10m height intervals using bilinear raster grid interpolation.